Indices and Index Laws in $\mathbb{Z}$

Note: indices can be referred to as exponents.

1. Evaluate 2^5.
2. Evaluate $(-3)^4$ and compare it with -3^4. Explain the difference.
3. Simplify using index laws and leave your answer in index form: $5^3 \times 5^4$.
4. Evaluate and express as a fraction: 7^{-2}.
5. Simplify: $\dfrac{4^6}{4^2}$.
6. Simplify using index laws: $(3^2)^3 \div 3^4$.
7. Evaluate the expression: $(-2)^3 \times (-2)^2$.
8. Simplify fully: $\dfrac{(-5)^7 \times (-5)^2}{(-5)^4}$.
9. A population of microorganisms doubles every hour. If the starting population is 2^3 organisms, write an expression using index notation for the population after 5 hours and evaluate it.
10. Evaluate the following combined expression: $3^0 + 2^{-3} \times 2^5 - (-2)^2$.

Solutions

1. **32**
   Explanation: Multiply the base 2 by itself 5 times: $2 \times 2 \times 2 \times 2 \times 2 = 32$.

2. **$(-3)^4 = 81$, whereas $-3^4 = -81$**
   Explanation: For $(-3)^4$, the negative sign is inside the base, so $(-3) \times (-3) \times (-3) \times (-3) = +81$. For -3^4, only the 4 applies to the 3, yielding $-(3 \times 3 \times 3 \times 3) = -81$.

3. **5^7**
   Explanation: Using the first law of indices ($a^m \times a^n = a^{m+n}$): $5^3 \times 5^4 = 5^{3+4} = 5^7$.

4. **$\frac{1}{49}$**
   Explanation: Using the negative index law ($a^{-n} = \frac{1}{a^n}$): $7^{-2} = \frac{1}{7^2} = \frac{1}{49}$.

5. **4^4 (or 256)**
   Explanation: Using the division law of indices ($\frac{a^m}{a^n} = a^{m-n}$): $\frac{4^6}{4^2} = 4^{6-2} = 4^4 = 256$.

6. **3^2 (or 9)**
   Explanation: Apply the power of a power rule ($(a^m)^n = a^{mn}$): $(3^2)^3 = 3^6$. Then apply the division rule: $3^6 \div 3^4 = 3^{6-4} = 3^2 = 9$.

7. **-32**
   Explanation: Evaluate each term or use index laws: $(-2)^3 \times (-2)^2 = (-2)^{3+2} = (-2)^5 = -32$.

8. **$-3,125$ (or $(-5)^5$)**
   Explanation: Combine the numerator using addition of exponents: $(-5)^7 \times (-5)^2 = (-5)^9$. Then divide by the denominator: $\frac{(-5)^9}{(-5)^4} = (-5)^{9-4} = (-5)^5 = -3,125$.

9. **$2^8 = 256$ organisms**
   Explanation: The initial population is 2^3. Doubling over 5 hours multiplies the population by 2^5. Using index laws: $2^3 \times 2^5 = 2^{3+5} = 2^8 = 256$ organisms.

10. **1**
    Explanation: Evaluate each term step-by-step:

$$\begin{aligned} 3^0 &= 1 \quad \text{(any non-zero number to the power of 0 is 1)} \\ 2^{-3} \times 2^5 &= 2^{-3+5} = 2^2 = 4 \\ (-2)^2 &= 4 \end{aligned}$$

    Substitute these back into the expression: $1 + 4 - 4 = 1$.